

MACHINE LEARNING TERMINOLOGY

CLASSIFIER

A program or a function which maps from unlabeled instances to classes is called a classifier.

CONFUSION MATRIX

A confusion matrix, also called a contingency table or error matrix, is used to visualize the performance of a classifier.

The columns of the matrix represent the instances of the predicted classes and the rows represent the instances of the actual class. (Note: It can be the other way around as well.)

Confusion Matrix

		Predicted classes	
		male	female
Actual classes	male	42	8
	female	18	32

ACCURACY (ERROR RATE)

Accuracy is a statistical measure which is defined as the quotient of correct predictions made by a classifier divided by the sum of predictions made by the classifier.

$$\text{accuracy} = (42 + 32) / (42 + 8 + 18 + 32)$$

which is 0.72

Let's assume we have a classifier, which always predicts "female". We have an accuracy of 50 % in this case.

Confusion Matrix

		Predicted classes	
		male	female
Actual classes	male	0	50
	Female	0	50

We will demonstrate the so-called accuracy paradox.

Confusion Matrix		Predicted classes	
		spam	ham
Actual classes	spam	4	1
	ham	4	91

The accuracy of this classifier is $(4 + 91) / 100$, i.e. 95 %.

The following classifier predicts solely "ham" and has the same accuracy.

Confusion Matrix		Predicted classes	
		spam	ham
Actual classes	spam	0	5
	ham	0	95

The accuracy of this classifier is 95%, even though it is not capable of recognizing any spam at all.

PRECISION AND RECALL

Confusion Matrix		Predicted classes	
		negative	positive
Actual classes	negative	TN	FP
	positive	FN	TP

The ideal spam filter would have 0 values for both FP and FN

ACCURACY IN CLASSIFICATION

We are interested in Machine Learning and accuracy is also used as a statistical measure. Accuracy is a statistical measure which is defined as the quotient of correct predictions (both True positives (TP) and True negatives (TN)) made by a classifier divided by the sum of all predictions made by the classifier, including False positives (FP) and False negatives (FN). Therefore, the formula for quantifying binary accuracy is:

Accuracy: $(TN + TP) / (TN + TP + FN + FP)$

PRECISION

Precision is the ratio of the correctly identified positive cases to all the predicted positive cases, i.e. the correctly and the incorrect cases predicted as positive. Precision is the fraction of retrieved documents that are relevant to the query. The formula:

Precision: $TP / (TP + FP)$

RECALL

Recall, also known as sensitivity, is the ratio of the correctly identified positive cases to all the actual positive cases, which is the sum of the "False Negatives" and "True Positives".

Recall: $TP / (TP + FN)$

F1-SCORE

f1-score best reflects the worst case scenario that the FN value is rising, i.e. ham is getting classified as spam!

$$F_1 = \frac{2}{\frac{1}{\text{recall}} + \frac{1}{\text{precision}}} = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$$

ACCURACY

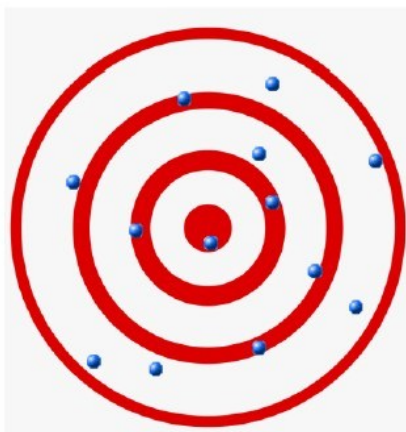
Accuracy is a measure for the closeness of the measurements to a specific value, while **precision** is the closeness of the measurements to each other, i.e. not necessarily to a specific value. To put it in other words: If we have a set of data points from repeated measurements of the same quantity, the set is said to be accurate if their average is close to the true value of the quantity being measured. On the other hand, we call the set to be precise, if the values are close to each other. The two concepts are independent of each other, which means that the set of data can be accurate, or precise, or both, or neither. We show this in the following diagram:



A: accurate and precise



B: precise, but not accurate



C: neither accurate nor precise



D: accurate, but not precise

SUPERVISED LEARNING

The machine learning program is both given the input data and the corresponding labelling. This means that the learn data has to be labelled by a human being beforehand.

UNSUPERVISED LEARNING

No labels are provided to the learning algorithm. The algorithm has to figure out the clustering of the input data.

REINFORCEMENT LEARNING

A computer program dynamically interacts with its environment. This means that the program receives positive and/or negative feedback to improve its performance.